

ICT502 – Internet and Web Development

Assessment 3 – Group Assignment

DUE DATE: Sunday 25 September 2026, 11:00 pm (AEST Melbourne time)

WEIGHTAGE: 25%

WORD COUNT: Report (1800 – 2000 Words) + Coding + Presentation.

Objective:

To provide students the opportunity to demonstrate their ability to understand and to apply the back end/front-end web development techniques to develop dynamic websites for given case study scenario. You will work in groups to design and develop a website that is both efficient and scalable, addressing a real-world problem scenario provided below.

Learning Outcomes:

Students should be able to demonstrate their achievements in the following unit learning outcomes:

ULO2: Design and implement responsive web pages and applications using HTML, CSS, and JavaScript.

ULO3: Apply professional web development practices in the design and construction of web applications.

ULO4: Design and develop web applications using appropriate front-end and back-end techniques.

ULO5: Assess security, privacy, accessibility, and usability considerations associated with web technologies, and communicate findings effectively in team environments.

Group Composition:

Groups should consist of 4 members. Collaboration is key. Divide tasks among members but ensure regular communication and consensus on design decisions.

Submission Guidelines:

The report is to be submitted in a word format. The assessment constitutes 30% of the total grade, with 20% allocated for the report and 10% for the presentation. The report should be between 1800 and 2000 words in length, excluding the title page and list of references. All pages must be numbered. Only one submission per group is needed. The similarity index of your report should be less than **20%**.

All students **MUST** submit **Three files**

Submit project files in a zipped folder and name the folder using the following

syntax: **[Group No]_[unitcode] _ [Campus].zip**

For example, 'Group A1_ICT502_MEL.zip'.

Submit your assessment report, name the document using the following

syntax: **[unitcode]_[Group No].doc (or '.docx')**

For example, 'ICT502_MEL ICT502_Group A1.doc'.

Submit at least one Meeting Minute, name the document using the following

syntax: **[Group No]_MM_#.doc (or '.docx')**

For example, 'Group A1_MM_1.doc'.

You must keep a backup copy of every assignment you submit until the marked assignment has been returned to you. In the unlikely event that one of your assignments is misplaced you will need to submit your backup copy.

Penalties for late submission:

Late submissions will incur a 10% deduction in marks for each day beyond the due date. Your PowerPoint file must be uploaded on the Moodle before the presentation. All students must give the presentation to receive marks. Zero marks will be awarded if a student is absent or fails to deliver the presentation.

Presentation:

Prepare a brief presentation summarizing your design process, coding, responsive, and functions of website, and key learning outcomes. Each group will give a presentation of 10 minutes (including 2-3 minutes of Q&A session) in the lecture of week 12.

Assignment Scenario:

Your team has been tasked with designing and developing a website for one of the scenarios listed according to your group number; **if you are in group N, you must choose scenario N**. For example, Group 10 must choose Scenario 10. If you work on a different scenario, your assignment will not be marked.

Scenario 1: Online Retail Store

Develop a fully functional e-commerce website where users can browse products by category, view detailed product descriptions, and add items to a shopping cart. The system should include user registration/login, a checkout process, and basic order confirmation. An admin interface should allow product management (add/edit/delete items).

Scenario 2: Restaurant Ordering System

Create a website that allows users to explore restaurant menus, filter items by category (e.g., vegetarian, drinks), and place online orders. The system should include order tracking, estimated delivery time, and a simple order history for registered users. Include a backend for restaurant owners to update menu items.

Scenario 3: E-Learning Platform

Design a web-based learning platform where users can register, enrol in courses, and access learning materials such as videos and documents. Include features like progress tracking, quizzes, and a dashboard for instructors to upload content and manage courses.

Scenario 4: Property Finder Website

Build a real estate platform where users can search for properties using filters such as location, price range, and property type. Each listing should include images, descriptions, and contact details. Include a form for users to submit inquiries and an admin panel to manage listings.

Scenario 5: Budget Tracker Application

Develop a web application that allows users to input income and expenses, categorise transactions, and view summaries through charts or tables. Include features such as budget setting, alerts for overspending, and the ability to export reports.

Scenario 6: Wellness Tracker System

Create a health monitoring website where users can log daily activities such as exercise, diet, and sleep patterns. The system should generate personalised tips based on input data and display trends over time using visual dashboards.

Scenario 7: Event Management Portal

Design a platform for creating and managing events. Users should be able to view upcoming events, register online, and purchase tickets. Organisers should have access to tools for managing registrations, tracking attendance, and promoting events.

Scenario 8: Career Opportunities Portal

Develop a job recruitment website where employers can post job listings and manage applications. Job seekers should be able to search for jobs, upload resumes, and apply online. Include filtering options and a dashboard for tracking applications.

Scenario 9: Customer Review Platform

Build a system that enables customers to submit feedback and rate services or products. The platform should display reviews, calculate average ratings, and generate reports for business owners. Include moderation features for managing inappropriate content.

Scenario 10: Trip Planner Application

Create a travel planning website where users can create itineraries, add destinations, and organise travel activities. Include options for saving plans, sharing itineraries, and integrating estimated costs for flights and accommodation.

Scenario 11: IT Services Company Website

Design a professional website for an IT services company, including pages for services, case studies, and client testimonials. Users should be able to submit project inquiries and receive estimated quotes based on selected services or technologies.

Scenario 12: Animal Adoption Platform

Develop a website for a pet adoption centre where users can browse available animals, filter by type or age, and submit adoption applications. Include a care cost calculator and an admin panel to manage listings and applications.

Scenario 13: Home Improvement Services Website

Create a website for a home improvement business (e.g., landscaping or renovations). Include a project cost estimator based on user inputs (e.g., size, materials), service descriptions, and a contact form for booking consultations.

Scenario 14: Digital Art Showcase

Design a website for an art gallery where artists can submit their work through an online form. Visitors should be able to browse collections, view artwork details, and submit inquiries. Include an admin interface for approving submissions.

Scenario 15: Local Sports Club Portal

Develop a website for a community sports club that allows users to register for teams, view schedules, and pay membership fees. Include a fee calculator based on age group or sport type, and a dashboard for administrators to manage teams and events.

Development Steps:

- **Design Layout and User Interface:** Create the web application's layout and user interface using HTML and CSS. This includes defining the overall structure of the web application, styling page elements, and ensuring a clear, consistent, and responsive design.
- **Logic and Functionality Implementation:** Implement the web application's logic and functionality using JavaScript and PHP. This may include form validation, database integration, user input handling, and server-side processing.
- **User Interaction and Database Design:** Develop a database system to manage user profiles, authentication, and page customisation. Use an SQL database and PHP to establish connectivity between the website and the database.
- **Security and Privacy Considerations:** Apply basic security and privacy practices in the web application. This may include secure login and authentication, password protection, input validation, protection against SQL injection, appropriate session handling, and responsible handling of user data.
- **Accessibility and Usability:** Ensure the web application is easy to use and accessible to different users. Consider clear navigation, readable text, appropriate colour contrast, meaningful labels, responsive design, keyboard accessibility, and a user-friendly interface.
- **Testing and Validation:** Thoroughly test the web application to ensure functionality, security, usability, and user-friendliness. Conduct both manual and automated testing where appropriate, including testing forms, links, database operations, login functions, responsiveness, and error handling.

- **Presentation Preparation:** Prepare a brief presentation summarising your design process, coding, responsiveness, security and privacy considerations, accessibility and usability features, website functions, and key learning outcomes. Each group will give a 10-minute presentation, including 2–3 minutes of Q&A, during the Week 12 lecture.

Submission Requirements:

Your submission should adhere to the following guidelines:

- Submit the source code in a zip file for the web application.
- Include a report containing the following:

Description of the web application and its features.

Discussion of the technologies utilised in the web application's development.

Overview of the design and development process.

Explanation of any challenges faced and their solutions.

Description of the testing process and its results.

Include the URL of the website in the Word document for review.

Please ensure that all submitted documents are well-organized and clearly presented.

General Assessment Requirement

Incomprehensible submissions:

Assessments provide the opportunity for students to demonstrate their knowledge and skills to achieve the required standard. To do this, assessment responses need to be both clear and easy to understand. If not, SISTC cannot determine that students have demonstrated their knowledge and skills. Assessments will, therefore, be marked accordingly, including the potential for 0 (zero) marks where relevant.

Check with marking rubric:

Before submitting your assessment, you should check it against the assessment criteria and the marking rubric included in this specification to ensure that you have satisfactorily addressed all the criteria that will be used to mark your submission.

Referencing

This unit requires you to use APA 7th edition system of referencing. See the Sydney International School of Technology and Commerce's quick reference guide – [APA 7th Referencing](#). It can be used in conjunction with the online tool Academic Writer - [https://extras.apa.org/apastyle/basics-7e/#/. \](https://extras.apa.org/apastyle/basics-7e/#/.)

Extensions

Extensions can only be granted for exceptional and/or unavoidable circumstances outside of your control. Requests for extensions must be made by 12 noon on the submission date using the online extension request form. All requests for extensions should be supported by appropriate evidence (e.g., a medical certificate in the case of ill health)

<https://forms.sistc.edu.au/232480027577862>

Feedback

Feedback will be supplied through Moodle. Authoritative results will be published on Moodle.

Academic Misconduct

To submit your assessment task, you must indicate that you have read, understood, and complied with, the Sydney International School of Technology and Commerce Academic Integrity and Student Plagiarism policies and procedures.

You must also agree that your work has not been outsourced and is entirely your own except where the work quoted is duly acknowledged. Additionally, you must agree that your work has not been submitted for assessment in any other course or program.

Use of Generative AI

In accordance with the College Academic Integrity Policy and the GenAI Traffic Light Threshold Colour coding, students may use Generative AI tools under specific conditions that promote ethical learning. Your submitted assignment should be your own work. It is NOT acceptable to find a generative AI tool to write your assignment. Never use generative AI to do the work that you are being assessed on. It is your responsibility to write and create your own assignments.

However, students may use Generative AI (GenAI) tools with caution and within the boundaries of academic integrity policies. The following uses are permitted only when not directly contributing to assessed work:

- As a learning aid to support understanding of concepts or content, but not for completing assessment tasks
- To explore unfamiliar topics as a starting point for research, provided that all referenced information in assessments is drawn from credible academic sources and appropriately cited

- To receive feedback on drafts or writing structure, with the expectation that students will critically evaluate and revise their work independently

All use of GenAI must align with the principles of academic integrity, and misuse may be considered a breach of assessment conditions. If you use generative AI to assist you to improve this assignment, it is necessary for you to acknowledge the use of generative AI appropriately. However, always ensure that your final submitted assessment is your own work, creation, and analysis.

Marking Rubric: A3 - Group Project Report

Performance Levels Criteria	Very Poor FNE (0-14%)	Poor F (15-49%)	Satisfactory P (50-64%)	Good C (65-74%)	Very Good D (75-84%)	Excellent HD (85-100%)
Scenario Understanding Scenario Understanding (1) Feature Definition (1). (Total: 2 marks)	No submission to this section / irrelevant content.	Fails to demonstrate understanding of the correct web application scenario or requirements, with major omissions.	Basic understanding of the web application's requirements, with significant gaps or omissions.	Good understanding of the web application's requirements but misses some important details or lacks depth.	Shows very good understanding of the web application's key requirements, with minor gaps in scenario integration.	Demonstrates thorough understanding of the web application's purpose, user's requirements. Integrates scenario context effectively.
Logic and Functionality Implementation (Total: 3 marks)	No submission to this section / irrelevant content	Inadequate use of the web technologies required for the web application and the adherence to best practices	Limited use of the web technologies required for the web application and the adherence to best practices	Satisfactory use of the web technologies required for the web application and the adherence to best practices	Proficient use of the web technologies required for the web application and the adherence to best practices	Exceptional use of the web technologies required for the web application and the adherence to best practices.
User Interaction and Database Design (Total: 3 marks)	No submission to this section / irrelevant content	No or broken API integration, making communication between front-end and back-end nonfunctional. Database integration is nonfunctional or highly insecure, with major performance issues.	API integration has significant issues, with poor data handling or broken functionality. Poor database integration, with performance issues, security risks, or data redundancy.	Satisfactory API integration but lacks efficiency or best practices for communication. Basic database integration, but queries are inefficient or prone to security issues.	Proficient API integration, with minor issues in data handling or performance. Good integration with minor inefficiencies in queries or database structure.	Well-implemented API integration, with efficient communication between front-end and back-end. Excellent integration with the database. Queries are efficient and data is well-structured.

Testing and Validation (Total: 2 marks)	No submission to this section / irrelevant content	No validation of testing results, or results are poorly documented.	Basic validation of results, with significant gaps in tracking and resolving issues.	Some results are validated, but documentation of outcomes is weak or unclear.	Most results are validated, but there may be gaps in evidence or minor issues unaddressed.	All results are thoroughly validated, with detailed evidence of testing outcomes. Issues are clearly tracked and addressed.
Security, Privacy, Accessibility and Usability (Total: 2 marks)	No submission to this section / irrelevant content.	Fails to adequately assess security, privacy, accessibility, or usability considerations. Recommendations are inappropriate, or unsupported.	Demonstrates a limited assessment of security, privacy, accessibility, and usability considerations, with notable omissions or limited justification.	Demonstrates a satisfactory assessment of key security, privacy, accessibility, and usability considerations. Recommendations are generally appropriate.	Demonstrates a detailed assessment of security, privacy, accessibility, and usability considerations, with only minor omissions. Provides appropriate recommendations.	Demonstrates a comprehensive assessment of security, privacy, accessibility, and usability considerations. Provides well-justified mitigation strategies and best-practice recommendations
Recommendations and team Contribution (2 marks)	No submission to this section / irrelevant content.	Recommendations are missing or not supported by the analysis. Demonstrates insufficient team contribution.	Recommendations unclear, or little supported by the analysis. Demonstrates limited team contribution.	Provides basic recommendations, although some may lack detail, justification, or relevance. Demonstrates satisfactory team contribution, with evidence of participation in group activities.	Provides relevant and practical recommendations supported by the findings. Demonstrates good team contribution, with evidence of consistent participation within the group.	Provides clear, insightful, and well-justified recommendations that are strongly aligned with the analysis. Demonstrates exceptional team contribution, with evidence of active collaboration.
Documentation and References (1 marks)	No submission to this section / irrelevant content.	Documentation lacks clarity, structure, or important details. Incorrect or incomplete use of APA 7 standard	Documentation lacks clarity, structure, or important details. Incorrect or incomplete use of APA 7 standard	Documentation is generally clear and organized but may lack some details. Correct use of APA 7 standard with few errors or omissions	Documentation is clear, well-organised, and covers all necessary aspects. Correct use of APA 7 standard with minor errors or omissions	Documentation is exceptionally clear, well-structured, and comprehensive. Correct and accurate use of APA 7 standard.

Presentation (Total: 10 marks) Content, structure and narrative (5). Clarity of the code demonstration (5).	Presentation was highly professional and demonstrated excellent presentation skills that kept the audience engaged and interested. Extremely well expressed with clear language and a linked and focused structured.	The presentation performance was acceptable and demonstrated some presentation skills. The audience were engaged throughout most of the presentation. Well expressed and a linked structure is evident.	The presentation was professional and demonstrated good presentation skills. The audience were mostly engaged throughout the presentation. Mostly well expressed and structured.	The presentation performance was acceptable and demonstrated some presentation skills. The audience were engaged throughout most of the presentation. Language expression and acceptable structure detracted from the message.		Presentation performance was of a poor standard or lacked credibility. The expression and structure of your presentation needs improvement.
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